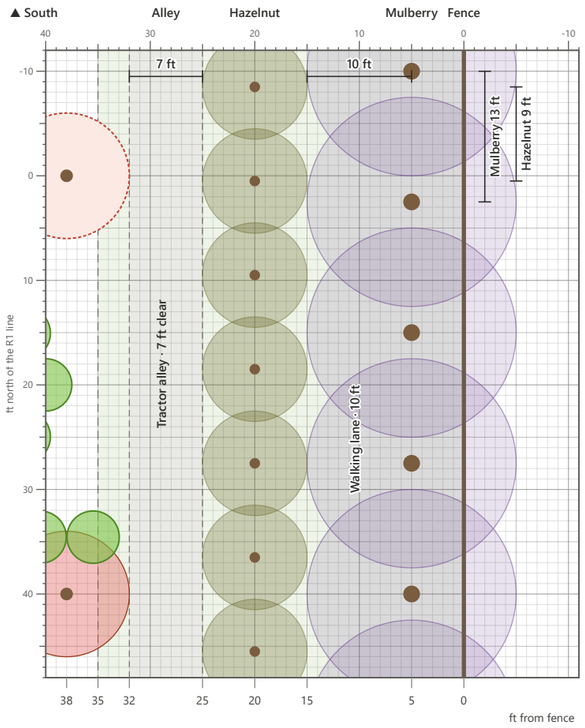
West Windbreak Guild Row — Options and Light Check

Three ways to arrange the mulberries and hazelnuts in one row, drawn beside the current plan, with the counts against the plants you have and a check on how much sun the hazelnuts get.

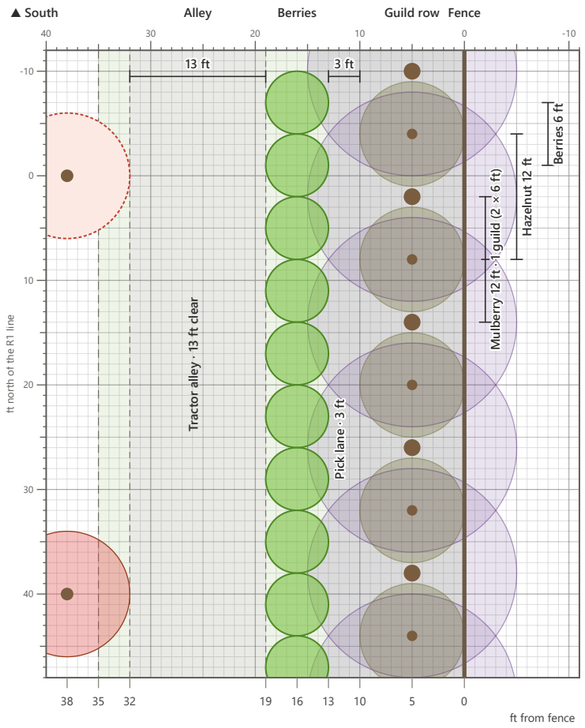
Mock-up only. Nothing here changes the planting plan or the data. The mock-up shows any one pattern in full: plan, profile, and the layers table.

The three patterns beside the current plan

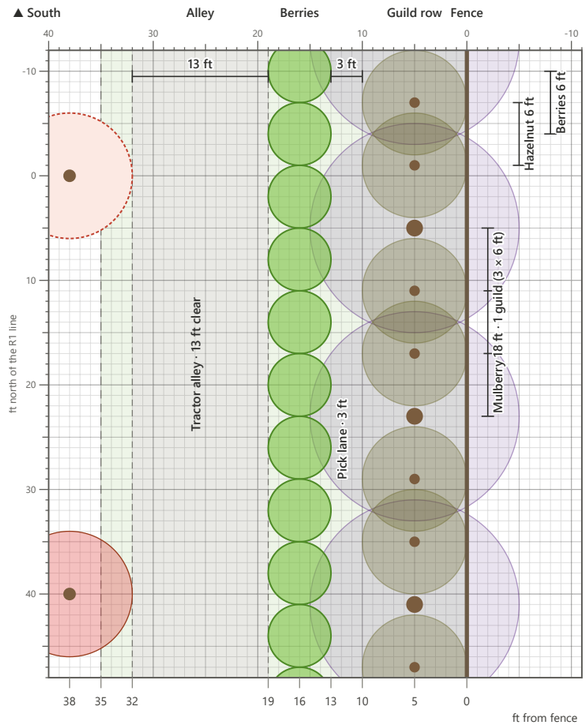
Current plan two rows, a 10 ft lane, a 7 ft alley



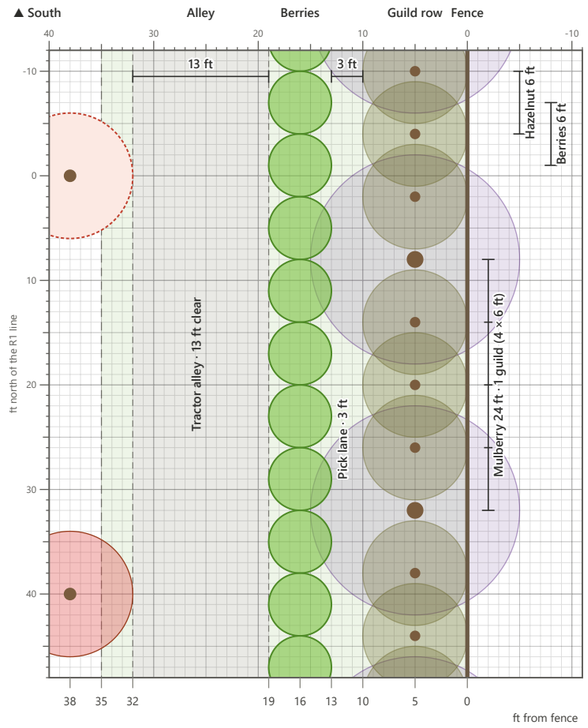
Pattern A: M–H one hazelnut between mulberries. Open in the mock-up



Pattern B: M–H–H two hazelnuts between mulberries. Open in the mock-up



Pattern C: M–H–H–H three hazelnuts between mulberries. Open in the mock-up



Red mulberry · Hazelnut · Small berries · First-column orchard tree and its limb room · A place for a plant still to add · Trunks: the large dot is a mulberry, the small dot a hazelnut

South up, west fence on the right, to scale: the same 60 ft of fence beside the first three orchard rows in every figure. The measurements west of the fence are the distance from one mulberry to the next, from one hazelnut to the next, and, in the proposed rows, from one berry bush to the next. In the proposed rows the mulberry line has a tick at every plant between its ends, so the gap from a mulberry to the hazelnut beside it reads off the same line. Every distance is in whole feet; where the current plan falls between two feet, the figure is rounded up. The row starts and ends with a mulberry, so it reaches 10 ft past R1 and past the last row in A, 13 ft in B, and 16 ft in C, against 10 ft for the mulberry row now.

Counts, space, and what you have on hand

	Current plan	A · M–H	B · M–H–H	C · M–H–H–H
THE ROW				
Pattern along the fence	two rows	M–H	M–H–H	M–H–H–H
Mulberry to mulberry	13 ft	12 ft	18 ft	24 ft
Hazelnut to hazelnut	9 ft	12 ft	6 ft	6 ft
PLANTS (ON HAND: 25 MULBERRIES, 34 HAZELNUTS, 41 BERRY BUSHES)				
Red mulberry	25	25 + 1 to add	18 7 spare	14 11 spare
Hazelnut	34	25 9 spare	34	34 + 5 to add
Small berries	41 around the C1 trees	41 + 9 to add in the front row	41 + 10 to add in the front row	41 + 11 to add in the front row
SPACE				
Strip used (fence to last row)	25 ft	19 ft	19 ft	19 ft
Tractor alley, clear	7 ft 6 ft at full width	13 ft 12 ft at full width	13 ft 12 ft at full width	13 ft 12 ft at full width
Lane between the rows	10 ft walking lane	3 ft pick lane	3 ft pick lane	3 ft pick lane

Pattern B, the two-hazelnut version, is outlined: it is the one the mock-up opens on. Spare plants can extend the row north or south past the orchard, where the plan says wind wraps around the ends. A place "to add" is a plant you would still buy or propagate.

Light on the hazelnuts

Hazelnuts set a crop in the sun, and the guild row puts them under and beside the mulberry crowns. This table works out how much of the sun reaches each hazelnut once every plant is full size, for the two ways a mulberry can grow. The first is the plan as drawn: kept to 20 ft wide, limbs from 14 ft. The second is the open-grown habit the plan's notes warn about: 30 ft wide, limbs from 8 ft.

	Current plan	A · M–H	B · M–H–H	C · M–H–H–H
MULBERRIES KEPT AS THE PLAN ASSUMES (20 FT WIDE, LIMBS FROM 14 FT)				
Share of the season's direct sun on the hazelnuts	74% 73%–84% across the row	66% 66%–71% across the row	70% 69%–72% across the row	75% 70%–84% across the row
Hours on June 21 with at least half the sun	10.9	10.5	11.0	11.8
Share of the season's sun on the berry row	—	70%	73%	77%
MULBERRIES LEFT TO SPREAD (30 FT WIDE, LIMBS FROM 8 FT)				
Share of the season's direct sun on the hazelnuts	55% 55%–63% across the row	31% 31%–33% across the row	38% 36%–41% across the row	44% 36%–49% across the row
Hours on June 21 with at least half the sun	8.0	7.0	7.5	8.0
Share of the season's sun on the berry row	—	49%	52%	55%

- **Kept as planned,** the hazelnuts in pattern B get within a few points of the sun they get in two rows, and those in A within ten. Pattern C is as sunny as today or sunnier, because its mulberries stand farthest apart.
- **Left to spread,** the guild row costs the hazelnuts much more than two rows do, because the hazelnuts sit under the crowns instead of 15 ft away. Two rows lose sun too, but they start from a better place.
- **The berry row** keeps most of the season's sun in every pattern, so Josta and the other berries are not the problem.

How it is worked out

- The sun's path at the lot (43.4737° N) in solar time, every 15 minutes on ten dates from May 15 to September 30, the months both species are in leaf. June 21 has 14.5 hours of sun more than 3° above the horizon.
- Every plant is an ellipsoid of its managed width and height from the plan: mulberry 20 ft wide and 14–42 ft up, hazelnut 10 ft wide to about 12 ft, berries 6 ft wide to about 7 ft. A hazelnut is tested against every other crown within 120 ft along the row, from its own middle.
- A beam loses about 8% of its remaining light for each foot of crown it crosses, so about four-fifths is gone over 20 ft. The sun counts in proportion to its height, the beam on level ground. No clouds, no trees outside the strip, and mature size: in the first eight to ten years the hazelnuts outgrow the mulberries and get full sun.
- This is a planning estimate, not a measurement. Differences of a few points are within its error. The gap between the two ways of growing the mulberries is the result to trust.

Which pattern to pick

B, two hazelnuts between mulberries

It is the version you described, it keeps most of both the mulberries and the hazelnuts you have, and with firm pruning it leaves the hazelnuts within a few points of today's light. It is the one the mock-up opens on.

C, if the hazelnuts are the crop you want

The most light on the hazelnuts, and the best chance of keeping some when a mulberry gets away from you. It needs the fewest mulberries, 14 against the 25 you have, but 39 hazelnuts against the 34 you have, and it leaves the most mulberries spare for another fence.

A, only for the tallest wall

It keeps the mulberries about as close as they are today and puts a hazelnut between each pair, the densest and tallest barrier. It needs 26 mulberries against the 25 you have. The hazelnuts get the least light, and a spreading mulberry would shade them out.

Whichever you pick, the tractor alley is the same 13 ft clear instead of 7, and the strip ends 19 ft from the fence instead of 25. The pattern changes what grows in the strip, not how much room it takes.

If you adopt one: what would change

- **The data.** `WINDBREAK` in `data/orchard.json` holds two rows 15 ft apart. It would become a guild row and a berry row, and `C1_GUILDS` would change if the berries move out of the orchard. Then run `python tools\property_data.py build`.
- **The model.** `web/orchard_model.js` (`windbreakLayout`, `laneGeometry`, `guildLayout`) assumes two single-plant rows and refuses rows closer than the shade gap. A row that mixes two kinds needs a new shape.
- **The pages.** The windbreak figures, table, and notes on the orchard pollination grid, and the tests that check them.
- **The water plan.** Zones 7, 8, and 9 in `data/water_harvesting.json` name the hazelnut row, the walking lane, and the C1 guilds.
- **The income page.** `docs/income_plan.js` counts 25 mulberries and 34 hazelnuts.
- **These two pages.** They would be retired once the plan itself draws the guild row.

The current plan is read from `data/orchard.json` through the orchard model. The proposal's choices, a 6 ft step along the row, a 3 ft pick lane, and a 6 ft berry row for the 41 bushes now planned around the C1 trees, are in `windbreak_guild_model.js`, with the sun and shade calculation.